Q9

(b)

(I) The claim is impossible.

Any for 2-D vectors u & v,

$$Z = u \otimes v \quad \text{with}$$

$$Z_{i*n+j} = u_i v_j \quad \forall \; 0 \le i,j \le n$$

This results in n^2 individual entries

$\Rightarrow$ Outputting n^2 entries needs $\Omega(n^2)$ //.

(II) Only compute $u_i v_j$ when an arbitrary entry Z_{i*n+j} is queried

(III). Kronecker product $u \otimes v$ independently pairs every element of u with every element of v, thus having discrete pairwise products $u_i v_j$.

Polynomial convolution sums these pairwise products whose indices i, j add up to k. So while Kronecker product yields n^2 terms, convolution yields a smaller vector $\in \mathbb{R}^{2n-1}$

Ordinary polynomial convolution takes $O(n^2)$. FFT optimizes this to $O(n \log n)$ by using point-value representation and n-th roots of unity. $\Rightarrow$ simple point wise multiplication in $O(n)$ and inverse FFT is used for interpolation - to retrieve the combined coefficients in $O(n \log n)$